

555 Timer Astable Multivibrator (≈ 1 Hz) Design Calculations

Objective

Design and simulate a NE555 timer in astable mode to generate a low-frequency square wave.

Components

R1=22 k Ω

R2=100 k Ω

R3=220 Ω

C1=10 μ F

C2=0.01 μ F

Supply=5 V

Design Equation

$$f = 1.44 / ((R_A + 2R_B) \times C)$$

Using R_A=22 k Ω , R_B=100 k Ω and C=10 μ F:

f $\approx$ 0.65 Hz, T $\approx$ 1.54 s.

Observation

The oscilloscope shows a stable square-wave output. Practical measurements may vary slightly due to simulation models and tolerances.

Applications

LED blinkers, pulse generation, timing circuits, digital clock inputs and laboratory demonstrations.

Conclusion

The NE555 operates correctly in astable mode, producing a continuous square wave suitable for low-frequency timing applications.